Joint Control Barrier Function for Target Visibility and Thrust Deliverability

A camera-plane QP over the commanded inertial acceleration

Soft–Precise–Landing

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Scope. Visibility is an *attitude* constraint — the body tilt moves the field of view — but the attitude actually commanded is a byproduct of the desired inertial acceleration $\boldsymbol{I}_a = ({}^{\mathcal{I}}a_x, {}^{\mathcal{I}}a_y, {}^{\mathcal{I}}a_z)$ that the outer control loop produces. Rather than solving a tilt-only QP and separately deciding how much vertical thrust to spend, we solve one QP directly over $\boldsymbol{I}_a$ that jointly enforces (i) the target stays in the field of view and (ii) the commanded thrust stays within what the vehicle can actually deliver. Both constraints are naturally expressed in acceleration units, so a single variable and a single feasible set serve both. The coupling between a candidate acceleration and the resulting feature motion is still the **rotational part of the IBVS interaction matrix**, L_ω, evaluated at the measured image point — exact, *depth-independent*, no de-rotation or virtual frame required. The barrier uses image features only — scale- and depth-free (no β, no optic flow, no disturbance term).

Notation

Symbol	Meaning
${}^{\mathcal{C}}\hat{\boldsymbol{r}}_i,\ {}^{\mathcal{C}}\hat{\boldsymbol{r}} = (x, y)$	camera image feature points / centroid, tangent units, per axis k
$L_\omega({}^{\mathcal{C}}\hat{\boldsymbol{r}})$	rotational interaction matrix (roll/pitch columns), $\partial\,{}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}/\partial\boldsymbol{\omega}_{\mathrm{rp}}$
$\boldsymbol{\omega}_{\mathrm{rp}} = [\omega_\phi, \omega_\theta]$	body roll/pitch rate (the *rotation-axis* vector)
$M = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$	lean$\to$rotation map: $\boldsymbol{\omega} = M\boldsymbol{\theta}$ (lean $+x$ is a rotation about $+y$)
$\widetilde{L_\omega} = L_\omega M$	effective tilt coupling (orthogonal M, so row norms are preserved)
$\boldsymbol{I}_a = (I_{a,x}, I_{a,y}, I_{a,z})$	commanded inertial accel., gravity-subtracted; the QP's decision variable
$a_z = \lvert I_{a,z}\rvert$	commanded vertical acceleration magnitude
P	fixed rotation (yaw + mount) carrying $\boldsymbol{I}_{a,xy}/a_z$ into image-tilt axes
$\boldsymbol{\theta} = P\,\boldsymbol{I}_{a,xy}/a_z$	*derived* lean-direction vector, used only to build the commanded attitude
A_{cap}	vehicle's maximum achievable thrust-to-mass ratio (measured)
$\boldsymbol{d}$	exogenous (translation) feature drift, $\boldsymbol{d} = {}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\mathrm{obs}} - L_\omega\boldsymbol{\omega}_{\mathrm{rp}}^{\mathrm{curr}}$
$\mathcal{R}, f$	resolution, focal (px); $\boldsymbol{\varphi}_{\max} = \mathcal{R}/(2f)$ half-FoV (tangent), the FoV edge
$\delta, \dot{\delta}$	primary-marker half-spread (loom rate)
τ	drift look-ahead; h barrier; g gravitational acceleration

1 Barrier on the real camera plane

Per image axis k,

$$\boxed{h_k = \varphi_{\max,k} - \lvert {}^{\mathcal{C}}\hat{r}_k\rvert - \delta_k}. \tag{1}$$

$h_k \geq 0 \iff$ the marker (outermost feature point) is on the sensor. ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$ is the *measured* camera image point — nothing is de-rotated.

2 Tilt$\to$feature coupling, expressed in acceleration units

A camera rotation moves a feature point by the rotational part of the IBVS interaction matrix, a standard and **depth-independent** object evaluated at the measured point ${}^{\mathcal{C}}\hat{\boldsymbol{r}} = (x, y)$:

$$ {}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}} = L_\omega({}^{\mathcal{C}}\hat{\boldsymbol{r}})\,\boldsymbol{\omega}_{\mathrm{rp}} + \boldsymbol{d}, \qquad L_\omega = \begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}. \tag{2}$$

L_ω contains **no depth**. The drift is measured directly by stripping the rotational term from the observed feature velocity, $\boldsymbol{d} = {}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\mathrm{obs}} - L_\omega\boldsymbol{\omega}_{\mathrm{rp}}^{\mathrm{curr}}$ (using the true gyro rate), so it carries only genuine target motion, not our own tilting.

A commanded lean produces a rotation about the perpendicular axis, $\boldsymbol{\omega} = M\boldsymbol{\theta}$ (3), and a lean is itself the direction of the commanded lateral acceleration relative to the commanded vertical acceleration: $\boldsymbol{\theta} = P\,\boldsymbol{I}_{a,xy}/a_z$, where P folds together the yaw alignment and the fixed camera-mount rotation. Substituting both into (2) gives the feature velocity directly as a function of the acceleration command:

$$\boldsymbol{\omega} = M\boldsymbol{\theta}, \quad M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad L_\omega M\boldsymbol{\theta} = \underbrace{\left(\frac{L_\omega M\,P}{a_z}\right)}_{N(a_z)} \boldsymbol{I}_{a,xy}. \tag{3}$$

$N(a_z) = \widetilde{L_\omega} P/a_z$ is the box constraint's Jacobian with respect to the lateral acceleration directly. It depends on a_z — the same acceleration this QP also solves for — which is exactly the coupling the joint formulation below is built to resolve, rather than treating a_z as an external given. For a finite acceleration change the exact map is the rotation homography of the measured point; the first-order form

$$^{\mathcal{C}}\hat{\boldsymbol{r}}(\boldsymbol{I}_{a,xy}) \approx {}^{\mathcal{C}}\hat{\boldsymbol{r}} + N(a_z)\,\boldsymbol{I}_{a,xy} - \widetilde{L_\omega}\,\boldsymbol{\theta}_{\mathrm{curr}} \tag{4}$$

is the local linearization, accurate over one control step. $\boldsymbol{\theta}_{\mathrm{curr}}$ is read directly from the *realized* attitude (not from $\boldsymbol{I}_a$), so it needs no rotation from acceleration units.

3 The visibility–deliverability QP

Two physical constraints bind the same variable $\boldsymbol{I}_a$: the predicted feature position must stay inside the field-of-view box, and the commanded thrust vector must stay inside the vehicle's real envelope. Writing $\boldsymbol{I}_a^d$ for the unconstrained desired acceleration (the outer loop's own command, before any visibility consideration) and A_{cap} for the maximum achievable thrust-to-mass ratio,

$$\boxed{\begin{aligned} &\boldsymbol{I}_a^\star = \arg\min_{\boldsymbol{I}_a} \|\boldsymbol{I}_a - \boldsymbol{I}_a^d\|^2 \\ &\text{s.t.} \quad \left|{}^{\mathcal{C}}\hat{r}_k + [N(a_z)\boldsymbol{I}_{a,xy} - \widetilde{L_\omega}\boldsymbol{\theta}_{\mathrm{curr}}]_k + \tau d_k\right| \le m_k \quad \forall k, \qquad \left\|\boldsymbol{I}_a + g\boldsymbol{e}_3\right\| \le A_{\mathrm{cap}} \end{aligned}} \tag{5}$$

The first constraint is the same FoV box as before, now written directly on $\boldsymbol{I}_{a,xy}$; the second is the true thrust-deliverability sphere in gravity-shifted acceleration space (the vehicle's total thrust vector is $m(\boldsymbol{I}_a + g\boldsymbol{e}_3)$, and $\|\boldsymbol{I}_a + g\boldsymbol{e}_3\| \le A_{\mathrm{cap}}$ is exactly the statement that this stays within the measured maximum). Both are convex for a fixed a_z; since $N(a_z)$ itself depends on a_z, the two projections are iterated together (alternating projection onto the box at the current a_z estimate, then onto the sphere, updating a_z for the next pass) until convergence. This is a genuine joint solve for the lateral and vertical acceleration together, not a lateral solve followed by a separate vertical adjustment: the vertical component the visibility constraint can "spend" and the vertical component actually available from the thrust ceiling are reconciled inside the same iteration.

- **Directional** — only the outward-binding box row is active; inward (recovery) and tangential accelerations are free. No strangling.
- **Envelope-aware** — the deliverability constraint is the vehicle's actual measured thrust ceiling, not an angle assumed at hover; it tightens or loosens correctly as a_z moves away from hover, because it is expressed on the full acceleration vector rather than on a lean angle computed *as if* a_z were fixed.
- **Forward-invariant** — the look-ahead shifts the box anchor by $\tau\boldsymbol{d}$, so the QP accounts for centroid drift over the next τ s in both phases (Sect. 4).
- **Bounded by construction** — because $\boldsymbol{I}_a^\star$ always satisfies the sphere constraint, the derived lean $\boldsymbol{\theta}^\star = P\,\boldsymbol{I}_{a,xy}^\star/a_z^\star$ used to build attitude (Sect. 8) is automatically bounded by the true deliverable cone at the solved $a_z^\star$; no separate, hover-assumed angular cap is needed.

4 Two-phase δ (central-marker convention)

The designated *central* marker is the target. The predicted margin m_k is phase-dependent:

Phase 1 — central marker decodes. $\delta_k = 0$, so $m_k = \varphi_{\max,k}$ (full half-FoV). The barrier is centroid-only: the QP box spans the entire image plane, and only fires if the *centroid* drifts toward the edge. The central marker is deliberately allowed to grow and overflow as the UAV closes in — its visibility *is* the guarantee; using its large half-extent would fire the barrier early. $\dot{\delta}$ is tracked but excluded from m_k in Phase 1.

Phase 2 — central marker overflows (decode-fail, hysteresis-gated against flicker $\Rightarrow$ touchdown range). δ_{eff} **ramps in** over a few frames from the visible-*set* half-extent ($\frac{1}{2}\text{ptp}_k$ over ALL visible markers' corners, so $|{}^{\mathcal{C}}\hat{\boldsymbol{r}}| + \delta_{\text{eff}}$ is the true outermost feature — not one marker):

$$m_k = \varphi_{\max,k} - \delta_{\text{eff},k} - \tau\dot{\delta}_{\text{eff},k}.$$

The ramp (not a step) keeps h from dropping discontinuously at engagement, but clamp the ramp gain so δ_{eff} never *lags* the measured $\frac{1}{2}\text{ptp}$ — a ramp slower than the fill under-protects. Within Phase 2 the visible markers themselves leave (a mild inner sawtooth); $\dot{\delta}$ is reset across those switches. In Phase 2 the sphere constraint of (5) still applies unchanged; the deliverability side of the joint solve does not depend on which phase the visibility side is in.

Implementation: hysteresis threshold = 3 consecutive decode-fails before Phase 2 activates; ramp completes over 5 control cycles ($\approx$120 ms at 42 Hz). Both $\delta_{\text{eff}} = \alpha\,\delta_{\text{ref}}$ and $\dot{\delta}_{\text{eff}} = \alpha\,\dot{\delta}_{\text{ref}}$ are scaled by the same ramp factor $\alpha \in [0, 1]$, where δ_{ref} and $\dot{\delta}_{\text{ref}}$ are the half-extent and loom rate measured at the last successful decode.

5 Feasibility floor and hand-off

The box constraint alone is **infeasible** iff $m_k < 0$ for some k (the marker fills the frame within τ). In Phase 1, $m_k = \varphi_{\max,k} > 0$ always, so the box is **always feasible** on its own while the central marker decodes. The sphere constraint is always feasible by construction (it is satisfied by $\boldsymbol{I}_a = \boldsymbol{0}$, i.e. zero net commanded acceleration). Genuine joint infeasibility — the visibility box demanding more lateral acceleration than the thrust ceiling can deliver at any achievable a_z — is handled by the iteration converging to the point on the sphere closest (in the box-projection sense) to feasibility, rather than by declaring failure; this is the same graceful degradation the alternating-projection method already provides for the box alone.

The perception-floor hand-off is triggered by *central-marker overflow* (Phase 2 activation, hysteresis-gated), not by box infeasibility. On Phase 2 entry, or if $m_k < 0$ under Phase 2 margins, raise the perception-floor flag and hand off to the texture-free ring flow.

6 Assumptions (and the practical challenge each carries)

1. **The coupling $\widetilde{L_\omega} = L_\omega M$ and its sign convention are calibrated.** The fixed 90° map $\boldsymbol{\omega} = M\boldsymbol{\theta}$ (3) bridges the lean direction and the rotation-axis rate that L_ω couples to. Any residual mount-dependent sign of L_ω is a one-time check: command a small acceleration and confirm the resulting $\Delta\,{}^{\mathcal{C}}\hat{\boldsymbol{r}}$ matches $\widetilde{L_\omega}$ in sign before trusting the barrier.

2. **The filtered acceleration is realized** (the SO(3) loop tracks the attitude built from $\boldsymbol{I}_a^\star$). *Challenge:* attitude bandwidth $\Rightarrow$ lag; the sphere constraint bounds magnitude, not delay.

3. **The drift $\boldsymbol{d}$ is the translation part** (${}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$) and $\dot{\delta}$ the loom. *Challenge:* finite differences $\Rightarrow$ filter + binding-axis hysteresis; the disturbance enters as *measured* motion — no separate $\boldsymbol{d}_h$ term. The homography centroid (hence ${}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\text{obs}}$) must stay *smooth across marker-set changes* (the homography re-fits on a different visible subset) or $\boldsymbol{d}$ spikes — filter / hold a stable subset across a switch.

4. **First-order L_ω over the cycle.** Exact is the rotation homography of ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$; the linearization (4) is accurate over one control step.

5. A_{cap} **is the vehicle's true achievable thrust-to-mass ratio**, measured rather than assumed — this replaces an earlier fixed-angle deliverability assumption with a direct, always-current physical bound.

7 Scale-freeness and depth-freeness

The barrier (1), the coupling (2), and the QP (5) contain only the measured image point ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$, δ, $\varphi_{\max} = \mathcal{R}/(2f)$, the body rate, the commanded acceleration, and the vehicle's own thrust-to-mass ratio A_{cap} and gravitational constant g. L_ω **is depth-independent by construction** (no $\mathcal{V}z_{\text{t}}$, no β), and A_{cap}, g are fixed vehicle/physical constants, not measurements of target depth or altitude — the whole formulation remains manifestly scale- and depth-free. The only metric quantity upstream is the geometric tracker's R_d (a controller output).

8 Implementation (per control cycle)

1. *(once)* Calibrate L_ω's sign / axis convention (Assumption 1); measure A_{cap}.

2. Measure ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$ (board centroid, homography). Advance the two-phase δ state: if central marker decodes, reset Phase 2 counters and set $\delta_{\text{eff}} = 0$; otherwise increment the decode-fail counter, and once the hysteresis threshold is reached, ramp δ_{eff} toward the visible markers' $\frac{1}{2}$ptp.

3. Evaluate $L_\omega({}^{\mathcal{C}}\hat{\boldsymbol{r}})$; form the drift $\boldsymbol{d} = {}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$ and $\dot{\delta}_{\text{eff}}$ (filtered; zero in Phase 1). Read $\boldsymbol{\theta}_{\text{curr}}$ from the current realized attitude.

4. Two-phase predicted margin: Phase 1: $m_k = \varphi_{\max,k}$; Phase 2: $m_k = \varphi_{\max,k} - \delta_{\text{eff},k} - \tau\dot{\delta}_{\text{eff},k}$. If $m_k \leq 0$ for any k, raise the perception-floor flag and hand off to the ring estimator. Otherwise build the joint QP (5).

5. Starting from $\boldsymbol{I}_a^d$ (the outer loop's unconstrained desired acceleration), jointly project onto the FoV box (using $N(a_z)$ at the current a_z estimate) and the deliverability sphere, iterating until convergence, to obtain $\boldsymbol{I}_a^\star$.

6. **Build the desired attitude directly from $\boldsymbol{I}_a^\star$.** Recover $\boldsymbol{\theta}^\star = P\,\boldsymbol{I}_{a,xy}^\star / a_z^\star$ and set the desired body-z axis $\boldsymbol{r}_{d3} = \big[-R_z(\psi)\boldsymbol{\theta}^\star,\, 1\big]/\big\|[\,\cdot\,, 1]\big\|$, passing R_d to the SO(3) inner loop and taking the thrust *magnitude* from $\boldsymbol{I}_a^\star$ directly. Building attitude from the un-filtered $\boldsymbol{I}_a^\star$ (rather than from a separately low-pass-filtered acceleration command) keeps the barrier's guarantee

landing exactly on the commanded attitude, with no smearing from a downstream filter time constant.